

Closed-Form Enumeration of Edge-Coloured Linear Snake Polyominoes

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Abstract

We enumerate edge-coloured linear snake polyominoes: rows of L unit cells, each cell carrying a $\{0, 1\}$ -colouring of its four edges subject to a local balance condition, counted up to a group of order 32 generated by the plane symmetries of the strip, reversal of the row, and global exchange of the two colours. Writing N_L for the resulting count, we prove that $(N_L)_{L \geq 2}$ satisfies a linear recurrence with constant integer coefficients of minimal order 12, whose characteristic polynomial factors over $\mathbb{Q}$ as

$$P(x) = (x - 3)(x - 1)(x^2 - 3)(x^2 - 5x - 5)(x^2 - 3x + 1)(x^4 - 5x^2 - 5).$$

Solving the associated Vandermonde system yields the exact partial-fraction representation of N_L ; the dominant eigenvalue is $\mu = \frac{1}{2}(5 + 3\sqrt{5})$, the larger root of $x^2 - 5x - 5$, with amplitude $\frac{1}{20}(5 + 2\sqrt{5})$. The subdominant real eigenvalues $\frac{1}{2}(3 \pm \sqrt{5}) = \varphi^{\pm 2}$ link the sequence to the Lucas numbers through $\varphi^{2L} + \varphi^{-2L} = L_{2L}$. The closed form is verified against exact enumeration for $1 \leq L \leq 1000$ without discrepancy, and a 300-digit reconstruction of the algebraic form agrees to more than 200 digits. The recurrence evaluates N_{1000} , an integer of 768 digits, in a few milliseconds. The sequence 4, 21, 109, 586, 3326, $\dots$ does not currently appear in the OEIS.

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1 Introduction

Polyominoes, the edge-connected unions of unit cells in the square lattice, are classical objects of combinatorics and statistical mechanics. Attaching a $\{0, 1\}$ colour to each edge of each cell produces families of coloured configurations whose counts grow exponentially in the number of cells, with growth rate governed by the spectrum of a transfer matrix. We study the simplest such family, the *linear snake*: a straight row of L cells.

Each cell carries a colouring of its four edges subject to a balance condition, and adjacent cells must agree on their shared edge. Two colourings are identified when related by a symmetry of the ambient strip, by reversal of the row, or by exchange of the two colours; the full symmetry group has order 32. Our aim is an exact closed form for the number N_L of colourings up to this group.

The results are as follows. After fixing the model (Section 2), we show that N_L is a fixed rational combination of fixed-point counts, each a bilinear form in powers of a small transfer matrix, and therefore satisfies a linear recurrence with constant coefficients (Section 3). In Section 4 we identify the minimal such recurrence by an exact-arithmetic Berlekamp–Massey computation and factor its characteristic polynomial completely over $\mathbb{Q}$ (Proposition 4.1). Section 5 solves for the residues and states the closed form (Theorem 5.1) and its asymptotics. Section 6 records the connection with the Fibonacci–Lucas numbers, Section 7 the numerical verification, and Section 8 the computational cost.

2 The model

2.1 Cells and admissible colourings

Each cell is a unit square whose four edges are labelled S, E, N, W and coloured in $\{0, 1\}$. A cell colouring is thus a quadruple $(s, e, n, w) \in \{0, 1\}^4$.

Definition 2.1 (Admissible cell). A cell colouring (s, e, n, w) is *admissible* if $s + e + n + w \leq 2$. There are exactly 11 admissible colourings: one with no edge coloured 1, four with one, and six with two.

Definition 2.2 (Linear snake). A *linear snake* of length L is a sequence $c_1, \dots, c_L$ of admissible cell colourings such that the E-edge of c_i equals the W-edge of c_{i+1} for $1 \leq i < L$ (the *hinge* condition).

Figure 1 shows representative snakes for small L ; blue and red denote the colours 0 and 1.

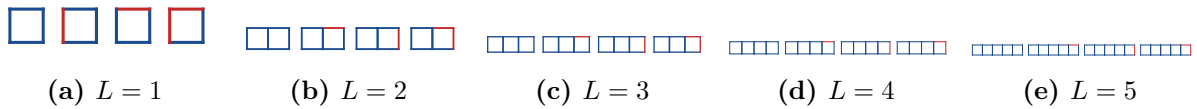


Figure 1: Representative colourings for $1 \leq L \leq 5$; blue is the colour 0, red the colour 1. The counts N_L are 4, 21, 109, 586, 3326.

2.2 The symmetry group

Definition 2.3 (Symmetry group). Let $G = D_4 \times \langle \iota \rangle \times \langle \phi \rangle$, of order $8 \cdot 2 \cdot 2 = 32$, where D_4 acts on the plane (and hence permutes the four edge labels of each cell), ι reverses the order of the cells, and ϕ exchanges the two colours. Two snakes are identified if related by an element of G .

For $L \geq 2$, the elements of G that map the horizontal strip to itself form a subgroup G_{eff} of order 16; the effective action on colourings is generated by the reflection σ_x in the long axis, the

half-turn combined with reversal $r_\pi\iota$, and the colour exchange ϕ . For $L = 1$ the reversal is trivial and all eight elements of D_4 act, so N_1 is a boundary value not governed by the recurrence derived below.

3 Transfer matrices and the counting sequence

3.1 The raw count

Ignoring symmetry, a snake is determined by choosing an admissible colouring for each cell subject to the hinge condition; the shared-edge colour is the natural transfer variable.

Proposition 3.1. *Let*

$$M = \begin{pmatrix} 4 & 3 \\ 3 & 1 \end{pmatrix},$$

where $M_{a,b}$ is the number of admissible cell colourings with $W = a$ and $E = b$. Then the number of linear snakes of length L , before any identification, is $\mathbf{1}^\top M^L \mathbf{1}$ with $\mathbf{1} = (1, 1)^\top$; the first values are 11, 65, 380, 2225, ... The eigenvalues of M are the roots of $x^2 - 5x - 5$, namely $\mu_{1,2} = \frac{1}{2}(5 \pm 3\sqrt{5})$.

Proof. For a single cell with fixed $W = a$ and $E = b$ the two remaining edges satisfy $s+n \leq 2-a-b$, giving $M_{a,b}$ admissible choices: $M_{0,0} = 4$, $M_{0,1} = M_{1,0} = 3$, $M_{1,1} = 1$. Concatenating L cells with matching hinges multiplies the transfer matrices, and summing over the two free boundary colours gives $\mathbf{1}^\top M^L \mathbf{1}$. The characteristic polynomial of M is $x^2 - 5x - 5$, with roots $\frac{1}{2}(5 \pm 3\sqrt{5})$. $\square$

The larger eigenvalue $\mu_1 = \frac{1}{2}(5 + 3\sqrt{5}) \approx 5.854$ is the growth rate of the raw count, and, as we show below, of the symmetry-reduced count N_L as well.

3.2 Existence of a linear recurrence

The symmetry-reduced count is obtained from the raw count by Burnside averaging over G_{eff} , together with an inclusion–exclusion that accounts for the colour exchange ϕ . For each symmetry g , the number $C_g(L)$ of g -invariant snakes is itself a bilinear form $u_g^\top M_g^{\lfloor L/2 \rfloor} w_g$ (with a bounded boundary correction) in the powers of a transfer matrix M_g of size at most 4; the halving of the exponent reflects that an invariant snake is determined by one half of its cells. Explicit matrices M_g for the elements of G_{eff} are given in the accompanying source [6].

Proposition 3.2. *There is a monic polynomial $Q \in \mathbb{Z}[x]$ such that $(N_L)_{L \geq 2}$ satisfies the linear recurrence with characteristic polynomial Q .*

Proof. Each $C_g(L)$ is a fixed rational combination of L -th powers of the eigenvalues of M_g , by Proposition 3.1 applied to M_g and the spectral decomposition. The count N_L is a fixed rational combination of the $C_g(L)$, hence a finite $\mathbb{Q}$ -linear combination $\sum_i A_i r_i^L$ in which the r_i range over the (finite) union of the spectra of the M_g . Any such combination satisfies the linear recurrence whose characteristic polynomial is $\prod_i (x - r_i)$; taking Q to be the least common multiple of the minimal polynomials of the M_g over $\mathbb{Q}$ gives an integer polynomial with the stated property. $\square$

Proposition 3.2 guarantees a recurrence but not its minimal order or explicit coefficients; we determine these next.

4 The minimal recurrence

We compute exact values $N_1, \dots, N_{100}$ by Burnside evaluation and apply the Berlekamp–Massey algorithm over $\mathbb{Q}$. Exact rational arithmetic is essential: it returns integer recurrence coefficients without the identification ambiguities of a floating-point fit.

Proposition 4.1. *The sequence $(N_L)_{L \geq 2}$ satisfies a linear recurrence of minimal order 12 with integer coefficients, namely, for $L \geq 14$,*

$$\begin{aligned} N_L = & 12N_{L-1} - 38N_{L-2} - 38N_{L-3} + 370N_{L-4} - 394N_{L-5} \\ & - 556N_{L-6} + 1160N_{L-7} - 690N_{L-8} + 370N_{L-9} \\ & + 330N_{L-10} - 750N_{L-11} + 225N_{L-12}, \end{aligned} \quad (4.1)$$

with seed values $N_2, \dots, N_{13}$. Its characteristic polynomial factors over $\mathbb{Q}$ as

$$P(x) = (x-3)(x-1)(x^2-3)(x^2-5x-5)(x^2-3x+1)(x^4-5x^2-5). \quad (4.2)$$

Adjoining the boundary value N_1 raises the minimal order to 13, the extra factor being x .

Proof. Berlekamp–Massey over $\mathbb{Q}$ applied to $N_2, \dots, N_{100}$ returns the order-12 recurrence (4.1); minimality is part of the algorithm’s output, no shorter recurrence being consistent with the data. Factoring the characteristic polynomial over $\mathbb{Q}$ gives (4.2). Finally, substituting the exact seed values and iterating (4.1) reproduces $N_{14}, \dots, N_{100}$ exactly, confirming the recurrence on the available data. $\square$

The factor $x^2 - 5x - 5$ in (4.2) is precisely the characteristic polynomial of the raw-count transfer matrix M of Proposition 3.1; it carries the dominant growth. The remaining factors arise from the fixed-point transfer matrices of the non-identity symmetries. In particular the biquadratic factor $x^4 - 5x^2 - 5$ equals $y^2 - 5y - 5$ evaluated at $y = x^2$, so its roots are the square roots of $\mu_{1,2}$; this square-root structure is the algebraic trace of the exponent-halving noted after Proposition 3.1.

Table 1: The twelve roots of $P(x)$, grouped by irreducible factor. Here $\varphi = \frac{1}{2}(1 + \sqrt{5})$ is the golden ratio, and $\mu_2 = \frac{1}{2}(5 - 3\sqrt{5}) < 0$.

Factor	Roots	Numerical values
$x-3$	3	3
$x-1$	1	1
x^2-3	$\pm\sqrt{3}$	± 1.732
x^2-5x-5	$\mu_{1,2} = \frac{1}{2}(5 \pm 3\sqrt{5})$	5.854, -0.854
x^2-3x+1	$\nu_{1,2} = \frac{1}{2}(3 \pm \sqrt{5}) = \varphi^{\pm 2}$	2.618, 0.382
x^4-5x^2-5	$\pm\sqrt{\mu_1}, \pm i\sqrt{-\mu_2}$	$\pm 2.420, \pm 0.924i$

5 The closed form

Since the roots of P are simple, $(N_L)_{L \geq 2}$ has a unique representation $N_L = \sum_{i=1}^{12} A_i r_i^L$. Imposing this at $L = 2, \dots, 13$ gives a 12×12 Vandermonde system with the roots of P ; its exact solution yields the residues in Table 2.

Theorem 5.1. *For all $L \geq 2$,*

$$\begin{aligned} N_L = & -\frac{1}{4}\left(3^L + 1 + (\sqrt{3})^L + (-\sqrt{3})^L\right) \\ & + \frac{1}{20}(5 + 2\sqrt{5})(\mu_1^L + \nu_1^L) + \frac{1}{20}(5 - 2\sqrt{5})(\mu_2^L + \nu_2^L) + G(L), \end{aligned} \quad (5.1)$$

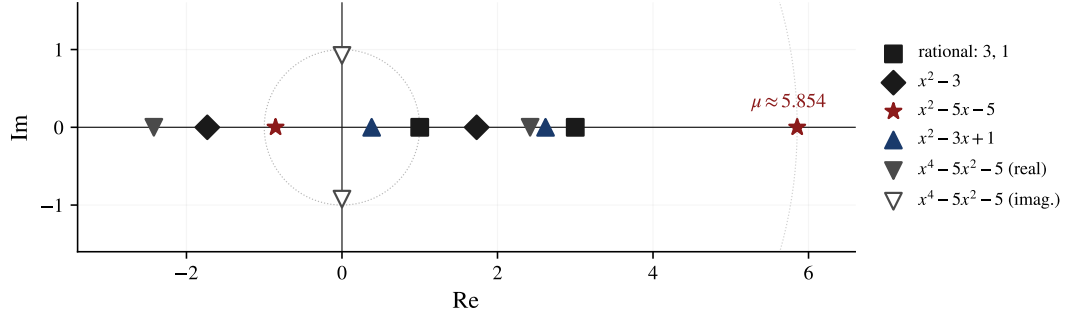


Figure 2: The twelve roots of $P(x)$ in the complex plane, grouped by irreducible factor. The dominant root $\mu_1 \approx 5.854$ governs the growth of N_L ; the two purely imaginary roots come from $x^4 - 5x^2 - 5$. The dotted circle has radius μ_1 .

Table 2: Residues A_i in the representation $N_L = \sum_i A_i r_i^L$. The residues for $\pm\sqrt{\mu_1}$ and $\pm i\sqrt{-\mu_2}$ are algebraic numbers of a longer form; their numerical values are given.

Root	Residue A_i	Numerical
3	$-\frac{1}{4}$	-0.250000
1	$-\frac{1}{4}$	-0.250000
$\pm\sqrt{3}$	$-\frac{1}{4}$	-0.250000
μ_1	$\frac{1}{20}(5 + 2\sqrt{5})$	0.473607
μ_2	$\frac{1}{20}(5 - 2\sqrt{5})$	0.026393
ν_1	$\frac{1}{20}(5 + 2\sqrt{5})$	0.473607
ν_2	$\frac{1}{20}(5 - 2\sqrt{5})$	0.026393
$+\sqrt{\mu_1}$	algebraic	0.911303
$-\sqrt{\mu_1}$	algebraic	0.035911
$\pm i\sqrt{-\mu_2}$	$0.026393 \pm 0.063859 i$	

where $G(L)$ collects the contributions of the four roots of $x^4 - 5x^2 - 5$ with the residues of Table 2, and where $N_1 = 4$ is a boundary value. Consequently

$$N_L \sim A\mu^L, \quad A = \frac{1}{20}(5 + 2\sqrt{5}), \quad \mu = \frac{1}{2}(5 + 3\sqrt{5}), \quad (5.2)$$

with $A \approx 0.4736068$ and $\mu \approx 5.8541020$; the relative error is $1 + \mathcal{O}((3/\mu)^L) = 1 + \mathcal{O}(0.513^L)$.

Proof. Equation (5.1) is the partial-fraction representation $N_L = \sum_i A_i r_i^L$ with the residues of Table 2, grouped by irreducible factor; the equal residues on μ_1, ν_1 and on μ_2, ν_2 have been combined. For (5.2), $\mu_1 = \mu$ is the unique root of P of maximum modulus (Table 1), so $A\mu^L$ dominates. The root of second-largest modulus is the rational root $x = 3$ (of modulus 3, exceeding $\nu_1 \approx 2.618$ and $\sqrt{\mu_1} \approx 2.420$), so the leading correction has magnitude $\mathcal{O}((3/\mu)^L)$ with $3/\mu \approx 0.513$; the residue of μ is $A = \frac{1}{20}(5 + 2\sqrt{5})$. $\square$

The convergence (5.2) is illustrated in Figure 3: the ratio $N_L/(A\mu^L)$ reaches 1 to floating precision by $L \approx 30$, and $N_L/N_{L-1} \rightarrow \mu$.

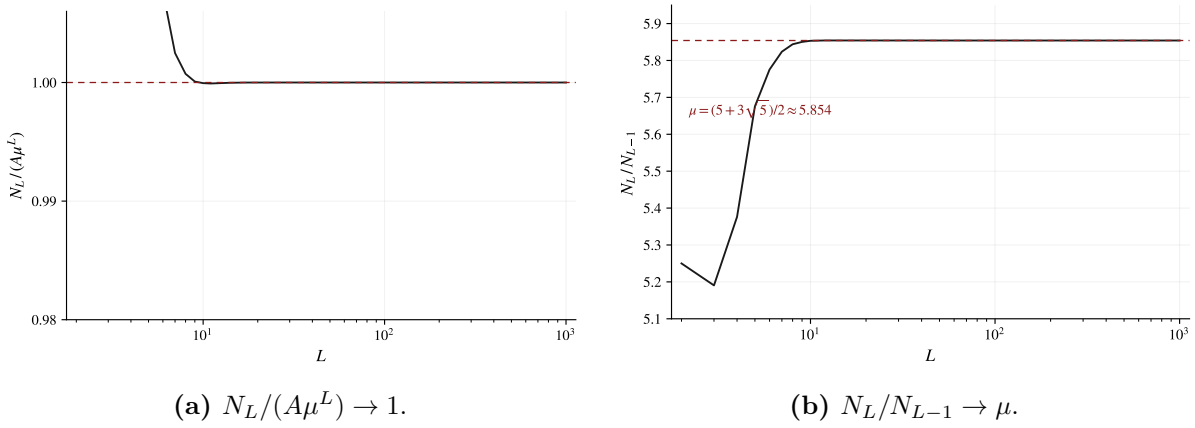


Figure 3: Convergence of the asymptotic (5.2), computed from the exact integer values of N_L for $L \leq 1000$; reference lines use the algebraic constants A and μ .

6 Arithmetic structure

The subdominant real roots are powers of the golden ratio, which ties the sequence to the Lucas numbers L_k (defined by $L_0 = 2$, $L_1 = 1$, $L_k = L_{k-1} + L_{k-2}$).

Proposition 6.1. With $\nu_{1,2} = \varphi^{\pm 2}$ and $\mu_{1,2} = \frac{1}{2}(5 \pm 3\sqrt{5})$,

$$\nu_1^L + \nu_2^L = L_{2L}, \quad \mu_1^L + \mu_2^L = a_L, \quad a_L = 5a_{L-1} + 5a_{L-2}, \quad (a_0, a_1) = (2, 5).$$

Proof. Since φ and $\psi = -\varphi^{-1}$ are the roots of $x^2 - x - 1$, $L_k = \varphi^k + \psi^k$; taking $k = 2L$ and using $\psi^2 = \varphi^{-2}$ gives $L_{2L} = \varphi^{2L} + \varphi^{-2L} = \nu_1^L + \nu_2^L$. For the second identity, $\mu_{1,2}$ are the roots of $x^2 - 5x - 5$, so $\mu_j^L = 5\mu_j^{L-1} + 5\mu_j^{L-2}$; summing over j gives the stated recurrence with $a_0 = \mu_1^0 + \mu_2^0 = 2$ and $a_1 = \mu_1 + \mu_2 = 5$. $\square$

Thus the “golden” part of (5.1) may be written using L_{2L} ; we do not pursue a fully Lucas-based form, since the integer recurrence (4.1) is already the most economical route to exact values.

7 Verification

Exact values $N_1, \dots, N_{1000}$ were produced by two independent computations: Burnside evaluation with transfer-matrix powers, and iteration of the recurrence (4.1) from the seed values. The two agree at all 1000 indices. Independently, the algebraic form (5.1) was evaluated in 300-digit floating-point arithmetic, with residues obtained by solving the Vandermonde system numerically; the result matched the exact integers to more than 200 digits for every $L \leq 100$, the residual being consistent with rounding at that precision.

Figure 4 shows that the decimal length of N_L grows along the line $L \log_{10} \mu$, as predicted by (5.2). Table 3 lists the first fifteen values.

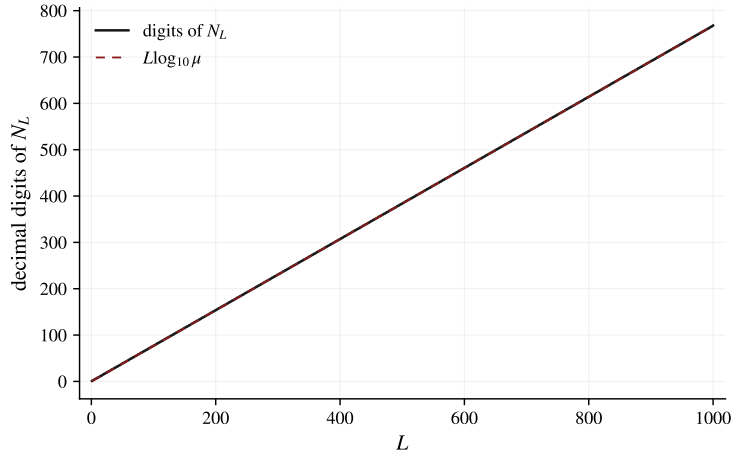


Figure 4: Decimal digits of N_L for $L \leq 1000$ against the line $L \log_{10} \mu$, with $\mu = \frac{1}{2}(5 + 3\sqrt{5})$.

Table 3: The first fifteen values of N_L , obtained identically by Burnside evaluation and by the recurrence (4.1).

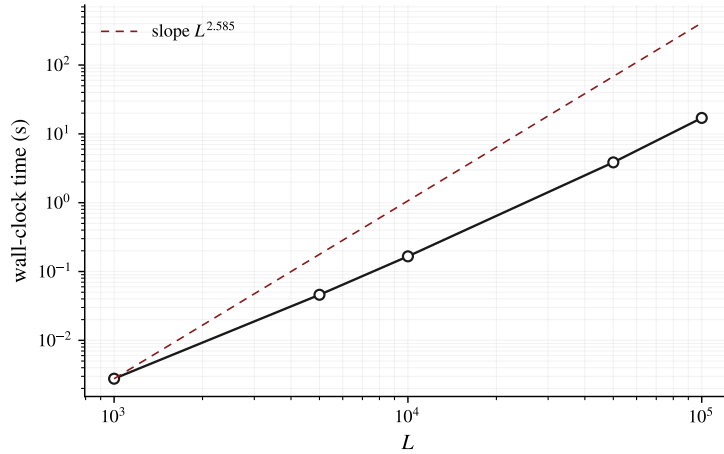
L	N_L	L	N_L
1	4	9	3,824,678
2	21	10	22,387,074
3	109	11	131,052,313
4	586	12	767,211,817
5	3,326	13	4,491,420,695
6	19,209	14	26,293,679,325
7	111,871	15	153,927,402,355
8	653,758		

8 Computational aspects

The recurrence (4.1) advances the sequence by twelve big-integer multiplications and eleven additions per step, on operands of size $\mathcal{O}(L \log \mu)$ bits. Writing $M(k)$ for the cost of multiplying two k -bit integers, computing $N_1, \dots, N_L$ costs $\mathcal{O}(L M(L))$; with the Karatsuba arithmetic of the reference implementation, $M(k) = \mathcal{O}(k^{1.585})$, giving an empirical $\mathcal{O}(L^{2.585})$. Table 4 and Figure 5 report measured timings; the arithmetic is exact at every scale, so N_{100000} , an integer of 76,746 digits, is computed without loss of precision.

Table 4: Wall-clock timings of the recurrence (4.1) on a single core (CPython 3.14).

L	Digits of N_L	Time
10^3	768	3 ms
$5 \cdot 10^3$	3,837	46 ms
10^4	7,675	166 ms
$5 \cdot 10^4$	38,373	3.85 s
10^5	76,746	17.0 s

**Figure 5:** Wall-clock time to compute N_L by the recurrence, on a log-log scale, with the reference slope $L^{2.585}$.

9 Remarks

Relation to the OEIS. The sequence $N_1, N_2, \dots = 4, 21, 109, 586, 3326, \dots$ does not currently appear in the OEIS. It should not be confused with the number of snake *shapes* (filaments) on n cells, which is $1, 1, 1, 2, 3, 7, 13, \dots$ (OEIS A002013) and counts unlabelled geometric figures rather than edge-colourings.

Exact versus algebraic evaluation. The recurrence (4.1) returns exact integers and is the method of choice for computation; the algebraic form (5.1) is exact but requires high-precision arithmetic to realise an integer at large L . The two are complementary: the recurrence for values, the closed form for structure and asymptotics.

General snakes. The same transfer-matrix and Burnside framework applies to bent snakes and to other balance conditions; the counting sequence again satisfies a linear recurrence, whose characteristic polynomial and residues may be recovered by the exact-arithmetic Berlekamp–Massey and Vandermonde computations used here.

References

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- [4] S. W. Golomb, *Polyominoes: Puzzles, Patterns, Problems, and Packings*, 2nd ed., Princeton University Press, 1994.
- [5] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.
- [6] Reference implementation: `src/snakes/compute_components.py` and `src/analysis/closed_form.py`.